PRODUCT REQUIREMENT DOCUMENT

Project: GeoLogix AI — Geopolitical Supply Chain Risk Intelligence

PRD V1.0 • CONFIDENTIAL

1. EXECUTIVE SUMMARY

GeoLogix AI is an enterprise-grade geospatial risk intelligence platform designed to map, simulate, and mitigate geopolitical supply chain disruptions. By combining historical trade bottleneck data with real-time LLM-driven event intelligence, GeoLogix AI enables global enterprises and sovereign logistics planners to transition from reactive crisis management to predictive risk hedging.

> **Core Value Proposition**
> Traditional ERPs track *where inventory is*; GeoLogix AI predicts *where inventory will be stranded*. By overlaying live maritime choke-points, political instability indexes, and supplier reliance nodes onto an interactive geospatial canvas, decision-makers reduce resolution lead times from days to minutes.

2. THE "WHY" & PROBLEM STATEMENT

Global supply chains face unprecedented volatility driven by armed conflicts, climate-induced maritime restrictions, and unilateral trade sanction policies. Legacy Supply Chain Event Management (SCEM) tools rely on linear, deterministic models that fail in non-linear geopolitical crisis environments.

$9.6B / day	68%	>72 Hours
SUEZ BLOCKAGE COST	CHOKE-POINT VULNERABILITY	AVG RESPONSE LATENCY

Primary Vulnerabilities in Legacy Systems

Domain	Legacy Constraint	GeoLogix AI Paradigm Shift
Geopolitical Blindspots	Tracks ocean routes purely via AIS signals without political context.	Integrates real-time geopolitical conflict zones and regulatory changes.
Cascading Risks	Analyzes Tier-1 suppliers in isolation; misses Tier-N dependencies.	Visualizes multi-tier ripple effects via dynamic node-based mind maps.
Action Latency	Manual rerouting analysis takes days of inter-departmental consensus.	AI-generated rerouting scenarios and automated supplier drafts in seconds.

2. TARGET USER PERSONAS (CONTINUED)

GeoLogix AI serves key decision-makers across corporate risk management, logistics operations, and sovereign policy planning who require actionable geospatial intelligence.

1. Chief Risk Officer (CRO)
Goal: Protect margin stability and board accountability.
Pain Point: Blind spots in Tier-2/3 raw material origins during political shifts.

2. Global Logistics Director
Goal: Maintain OTIF (On-Time In-Full) delivery metrics.
Pain Point: Sudden canal closures or tariff changes forcing emergency air-freight expense.

3. Sovereign Logistics & Trade Planner
Goal: Secure national strategic reserves (energy, semiconductors, food staple lines).
Pain Point: Lack of real-time multi-modal bottleneck modeling across international maritime straits.

3. PRODUCT VISION & KEY PERFORMANCE INDICATORS

To become the definitive global operating system for trade resilience, bridging physical geography with conversational artificial intelligence.

Phase	Core Strategic Focus	Key Deliverables
MVP (Short-Term)	Interactive maps with historical choke-point overlays & Gemini analysis.	Google Maps JS integration, historical incident dataset, automated incident summaries.
Phase 2 (Mid-Term)	Visual mind maps and automated mitigation workflows.	React Flow cascade mapping, automated supplier communication generator, route cost calculator.
Phase 3 (Vision)	Autonomous predictive routing and real-time AIS anomaly engine.	Predictive bottleneck ML model, digital twin simulation for global freight paths.

Target User KPIs

- **Mean Time to Resolution (MTTR):** Reduce response execution from 72 hours to <2 hours.
- **Margin Preservation:** Minimize spot-freight price spikes by 25% via early rerouting triggers.
- **Route Vulnerability Score:** 100% auditability across active primary shipping routes.

4. FUNCTIONAL REQUIREMENTS

The functional architecture consists of four interconnected core modules delivering end-to-end intelligence and execution workflows.

FR-1: Geospatial Disruption Canvas (Google Maps Dashboard)

- **Route Overlay:** Render primary multimodal trade corridors with custom Polyline vector layers.
- **Incident Hotspots:** Interactive markers showing historic and live disruption events (e.g., Red Sea attacks, Panama Canal drought).
- **Choke-Point Analytics:** Heatmaps and geo-fenced buffers surrounding strategic straits (Bab-el-Mandeb, Strait of Malacca, Suez).

FR-2: AI Disruption Intelligence Engine

- **On-Click Analysis:** Instant side-panel breakdown powered by Google Gemini API when clicking any incident marker.
- **Structured Output:** Automated generation of root causes, affected commodity codes (HS Codes), and estimated recovery timelines.
- **Geopolitical Sentiment:** AI scoring of regional stability indexes using continuous web context data.

FR-3: Visual Impact Mind Maps (Cascade Analysis)

- **Dependency Graphing:** Dynamic React Flow node visualizer showing first-order, second-order, and third-order fallout.
- **Node Lineage:** E.g., *[Zone A Conflict] → [Strait Closure] → [Port Congestion] → [Automotive Plant Shutdown]*.
- **Interactive Drill-Down:** Clicking graph nodes highlights specific impacted purchase orders and supplier contracts.

FR-4: Automated AI Risk Workflows

- **Alternative Routing:** Automated calculation of bypass routes (e.g., Cape of Good Hope detour) with ETA and fuel cost delta.
- **Dynamic Risk Scoring:** 0-100 vulnerability indexing based on route exposure time.
- **Supplier Notice Generator:** One-click draft generation of contractual force majeure notifications and re-routing directives.

5. NON-FUNCTIONAL REQUIREMENTS

To support enterprise adoption, the system must meet strict parameters regarding security, performance, and data scalability.

Category	Requirement Specification	Success Target
Security & Privacy	AES-256 encryption at rest; TLS 1.3 in transit. Enterprise tenant data isolation for custom shipping routes.	SOC 2 Type II compliance readiness; zero user-route data leak across tenants.
Performance & Latency	Gemini API response streaming for rapid summary rendering. Map tile rendering overhead optimized.	Sub-1.5 second initial response time for LLM insights; 60 FPS map interaction.
Scalability	GeoJSON spatial data streaming with vector tiling for large vessel track histories.	Support 100,000+ simultaneous geo-spatial points without browser degradation.

6. PROPOSED TECHNICAL STACK

A modern, modular stack built for rapid geospatial processing and seamless AI orchestration.

Layer	Technology Selected	Implementation Purpose
Frontend	React (Vite), TailwindCSS, React Flow	Component interface, styling system, and node-based mind map graphics.
Mapping & Spatial	Google Maps JavaScript API, GeoJSON	Base maps, custom vector layers, route rendering, and spatial querying.
Backend API	Node.js (Fastify) / Python (FastAPI)	High-throughput REST API, geospatial calculations, and third-party webhooks.
AI Integration	Google Gemini API (via Google AI Studio)	Multimodal analysis, geopolitical summaries, and automated response generation.
Database	PostgreSQL with PostGIS extension	Relational storage with spatial indexing for fast geographic queries.

Approval & Sign-Off

Document Status: Approved for MVP Development Sprint

Lead Author: Principal Product Manager, GeoLogix AI Platform